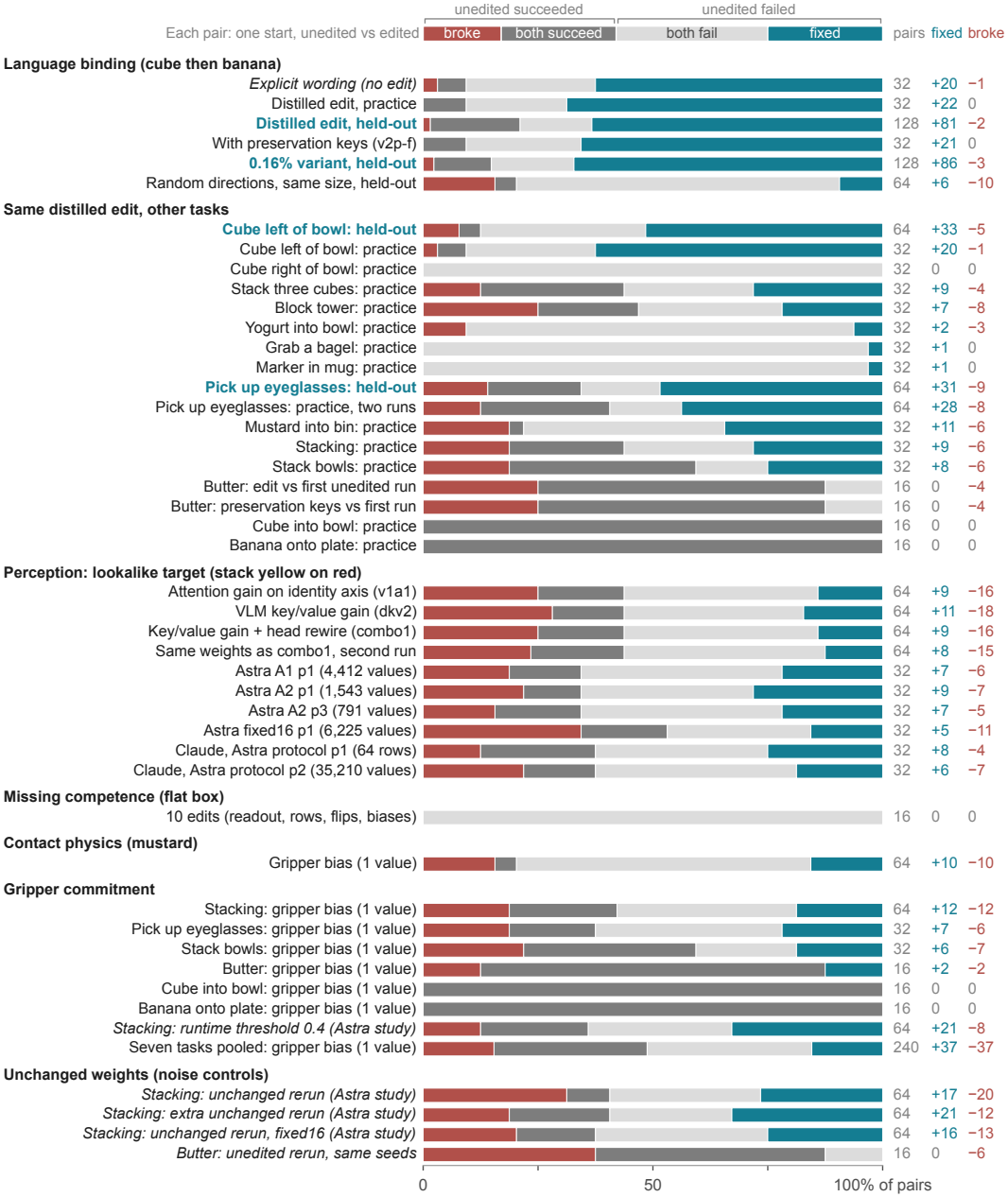

unedited succeeded
unedited failed
Each pair: one start, unedited vs edited
broke
both succeed
both fail
fixed
pairs
fixed
broke
Language binding (cube then banana)
Explicit wording (no edit)
32 +20 −1
Distilled edit, practice
32 +22 0
Distilled edit, held-out
128 +81 −2
With preservation keys (v2p-f)
32 +21 0
0.16% variant, held-out
128 +86 −3
Random directions, same size, held-out
64 +6 −10
Same distilled edit, other tasks
Cube left of bowl: held-out
64 +33 −5
Cube left of bowl: practice
32 +20 −1
Cube right of bowl: practice
32 0 0
Stack three cubes: practice
32 +9 −4
Block tower: practice
32 +7 −8
Yogurt into bowl: practice
32 +2 −3
Grab a bagel: practice
32 +1 0
Marker in mug: practice
32 +1 0
Pick up eyeglasses: held-out
64 +31 −9
Pick up eyeglasses: practice, two runs
64 +28 −8
Mustard into bin: practice
32 +11 −6
Stacking: practice
32 +9 −6
Stack bowls: practice
32 +8 −6
Butter: edit vs first unedited run
16 0 −4
Butter: preservation keys vs first run
16 0 −4
Cube into bowl: practice
16 0 0
Banana onto plate: practice
16 0 0
Perception: lookalike target (stack yellow on red)
Attention gain on identity axis (v1a1)
64 +9 −16
VLM key/value gain (dkv2)
64 +11 −18
Key/value gain + head rewire (combo1)
64 +9 −16
Same weights as combo1, second run
64 +8 −15
Astra A1 p1 (4,412 values)
32 +7 −6
Astra A2 p1 (1,543 values)
32 +9 −7
Astra A2 p3 (791 values)
32 +7 −5
Astra fixed16 p1 (6,225 values)
32 +5 −11
Claude, Astra protocol p1 (64 rows)
32 +8 −4
Claude, Astra protocol p2 (35,210 values)
32 +6 −7
Missing competence (flat box)
10 edits (readout, rows, flips, biases)
16 0 0
Contact physics (mustard)
Gripper bias (1 value)
64 +10 −10
Gripper commitment
Stacking: gripper bias (1 value)
64 +12 −12
Pick up eyeglasses: gripper bias (1 value)
32 +7 −6
Stack bowls: gripper bias (1 value)
32 +6 −7
Butter: gripper bias (1 value)
16 +2 −2
Cube into bowl: gripper bias (1 value)
16 0 0
Banana onto plate: gripper bias (1 value)
16 0 0
Stacking: runtime threshold 0.4 (Astra study)
64 +21 −8
Seven tasks pooled: gripper bias (1 value)
240 +37 −37
Unchanged weights (noise controls)
Stacking: unchanged rerun (Astra study)
64 +17 −20
Stacking: extra unchanged rerun (Astra study)
64 +21 −12
Stacking: unchanged rerun, fixed16 (Astra study)
64 +16 −13
Butter: unedited rerun, same seeds
16 0 −6
0
50
100% of pairs